

18. Sentro sa Pagsusuri, Pagsasanay at Pangasiwang Pang-agham at Teknolohiya Corp. (SENTROTEK)

Address : No. 208 Pilar St., Brgy. Addition Hills,
Mandaluyong City, 1550

Telephone No. : (02) 8721-6500, (02) 8721-9699,
(02) 8718-3514

Fax No. : (02) 8727-9016

E-mail : laboratory@sentrotek.com

Contact Person : Ms. Priscila P. Tongco
Laboratory Manager

Field of Testing : Chemical Testing and
Microbiological Testing

Chemical Testing

Product	Specific Test	Method Reference
I. Foods		
.01 Cereal and Cereal Products		
1. Breakfast Cereals	Protein	Kjeldahl, AOAC 2001.11/984.13
2. Cereal/cereal grains	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
3. Cultured seeds and grains	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
4. Soy flours concentrates and isolates	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
5. Flour, corn meal, corn grits, semolina	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
6. Frozen entrée containing rice or corn flour	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
7. Soy protein	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
8. Tofu	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition

Product	Specific Test	Method Reference
9. Pasta products and noodles (e.g. rice paper, rice vermicelli, soybean pastas and noodles)	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 969.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
	Caffeine	HPLC (AOAC 979.08)
10. Starch	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.02 Nuts and Nut Products		
1. Peanut butter and other nut butters	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)

	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.03 Dairy Products		
1. All cheese made from pasteurized milk (cottage cheese, soft & semi-solid cheese)	Protein	Kjeldahl, AOAC 2001.11/984.13
2. Processed cheese spread	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
3. Ice cream and sherbet plain and flavoured	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
4. Ice cream with added ingredients	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
5. Flavored ice	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)

	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
6. Milk powders (whole, nonfat or filled milk, buttermilk, whey & whey protein concentrates)	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
7. Sweetened condensed milk	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
8. Liquid Milk (evaporated or Ready-To-Drink) and Cream (Ultra-Heat Temperature/ Sterilized)	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
9. Pasteurized milk	Total Kjeldahl Nitrogen	AOAC 19 th edition

Product	Specific Test	Method Reference
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
10. Pasteurized cream	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
11. Yogurt and other fermented milk	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
.04 Meat and Meat Products		
1. Dried animal products	Protein	Kjeldahl, AOAC 2001.11/984.13
2. Meat paste and pate (heat treated)	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon

3. Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
4. Packed cooked cured/salted meat (ham, bacon)	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
5. Fermented comminuted meat, not cooked (dry and semi dry fermented sausages)	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
6. Cooked poultry meat, frozen to be re-heated before eating (e.g. prepared frozen meals)	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)

Product	Specific Test	Method Reference
7. Cooked poultry meat, frozen, ready- to-eat (e.g. turkey rolls)	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
8. Cured/smoked poultry meat	pH	AOAC 981.12
9. Dehydrated poultry products	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
10. Fresh/frozen raw chicken (during processing)	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
11. Meat products in hermetically sealed containers	Alcohol Content	In-house Method (APIL-FOOD-E001)
.05 Fish and fish products, including molluscs, crustaceans, and echinoderms		
1. Fresh and frozen fish and cold-smoked	Protein	Kjeldahl, AOAC 2001.11/984.13

	Fat	Ether extraction (AOAC 920.39C)Acid Hydrolysis (AOAC 922.06/948.15)
2. Pre-cooked breaded fish	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)

Product	Specific Test	Method Reference
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
3. Frozen cooked crustaceans	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
4. Cooked, chilled & frozen crabmeat	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
5. Fish and shellfish products in hermetically sealed containers	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
6. Smoked, dried, fermented, and/or salted fish and fish products	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
.06 Sugar and sugar products		

1. Refined and raw sugar	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)

Product	Specific Test	Method Reference
2. Brown sugar	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
3. Sugar solutions and syrups	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
4. Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
5. Honey	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
6. Table-top sweeteners, including those containing high- intensity sweeteners	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.07 Confectioneries		
1. Cocoa Powder	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
2. Chocolate products	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
3. Chocolate confectioneries (chocolate bars, blocks, bonbons)	Sodium Chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
4. Sugar confectioneries (hard and soft candies, toffees, caramel, fondants, creams, nougats and pastes)	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.08 Fruits, jams and other fruit products		
1. Frozen fruits	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
2. Coconut (desiccated)	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
3. Sun dried fruits	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
4. Jams, jellies, marmalades	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
5. Fruit-based spreads	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
6. Candied fruit	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
7. Fruit preparations (Pulp, purees, fruit toppings, and coconut milk)	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
8. Fermented fruit products	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
9. Fruit filling for pastries	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.09 Vegetable and vegetable products		
1. Frozen vegetables	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)

	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
2. Dried vegetables	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
3. Vegetable in vinegar, oil brine, or soybean sauce	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titratable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
4. Canned or bottled (pasteurized) or retort pouch vegetables	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
5. Fermented vegetable	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
6. Cooked or fried vegetables	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.10 Alcoholic Beverages		
1. Beer and Malt	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
2. Cider and perry	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)

	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
3. Grape wines	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
4. Wines other than grapes	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
5. Mead	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
6. Distilled spirits containing more than 15% alcohol	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
7. Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler- type beverages, low alcoholic refreshers	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.11 Soft drinks and cordial		
	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)

	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.12 Fruit juices, drinks and concentrates		
1. Fruit and vegetable juices	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
2. Fruit and vegetable nectars	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titratable Acidity	Titrimetry (AOAC 942.15)
3. Water-based flavoured drinks (Carbonated, Non-carbonated, Concentrates (liquid or solid))	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)

4. Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.13 Edible fats and oils		
1. Butter (whipped, pasteurized)	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
2. Butter made from unpasteurized milk and milk products	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
3. Fats and oils essentially free from water (vegetable oils and fats, lard tallow, fish oil and other animal fats)	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
4. Fat spread, dairy fat spreads and blended spread	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.14 Margarine		
	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C)Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titratable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.15 Eggs and eggs products		
1. Egg product (liquid, frozen, dried)	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
2. Preserved eggs, including alkaline, salted, and canned egg	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
3. Egg-based desserts (e.g. custard)	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.16 Bakery Products		
1. Frozen bakery products (ready to eat) with low-acid and high a _w fillings or toppings	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)

	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
2. Frozen bakery products (to be cooked) with low-acid and high a_w fillings or toppings (e.g. meat, pies, pizza)	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titratable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
3. Frozen and refrigerated dough (Chemically leavened)	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
4. Frozen and refrigerated doughs	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
5. Baked goods	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
6. Coated or filled, dried shelf-stable biscuits	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)

Product	Specific Test	Method Reference
.17 Antioxidants in foods		
	Benzoic acid	HPLC- In house Method
	Sorbic Acid	In-house Method (APIL-FOOD-S004)
.21 Vitamins in foods		
	Vitamins: A, B ₁ , B ₂ , B ₃ , B ₆ , C	HPLC- In house Validated Method
.24 Sauces, spices and condiments		
1. Dry mixes for soup and sauce	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)

	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
2. Yeast	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
3. Spices and herbs	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
4. Salad dressing (e.g. Mayonaise, Thousand island Mustard)	Titrateable Acidity	Titrimetry (AOAC 942.15)
5. Vinegars 6. Salt	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12

Product	Specific Test	Method Reference
7. Sauces and like products (e.g. fish sauce)	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
8. Soybean-based seasonings and condiments (e.g. soy sauce)	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
.25 Food supplement and/or Dietary Supplement		
	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)

	Total Sugars	Luff-Schoorl, Layne Eynon
	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)

Product	Specific Test	Method Reference
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
.26 Novel food products		
	Protein	Kjeldahl, AOAC 2001.11/984.13
	Fat	Ether extraction (AOAC 920.39C) Acid Hydrolysis (AOAC 922.06/948.15)
	Ash	Direct Method Furnace (AOAC 923.03/938.08/990.02)
	Total Sugars	Luff-Schoorl, Layne Eynon

	Total Dietary Fiber	Enzymatic Gravimetric (AOAC 991.43)
	Sodium, Potassium, Iron, Calcium	AAS (Modified AOAC 985.35/968.08)
	Other Minerals: Magnesium, Manganese, Zinc	AAS (AOAC 985.35)
	Heavy Metals: Chromium	AAS/ICP-AES USEPA 6010
	Other Tests: Histamine	Fluorometry (AOAC 979.08)

Product	Specific Test	Method Reference
	Titrateable Acidity	Titrimetry (AOAC 942.15)
	Free Fatty Acid	Titrimetry (AOAC 940.28)
	Peroxide Value	Titrimetry (AOAC 965.33)
	Carbohydrate Energy	Food Analysis 2 nd edition
	Sodium chloride	AOAC 960.29
	pH	AOAC 981.12
	Total Kjeldahl Nitrogen	AOAC 19 th edition
	Caffeine	HPLC (AOAC 979.08)
	Cholesterol	GC (AOAC 994.10)
	Saturated Fat, Transfat, Fatty Acid Composition	GC (AOAC 960.33, AOAC 996.06)
	Sucrose, Glucose, Maltose, Lactose	HPLC-In House Validated Method
	Fructose	In house Method (APIL-FOOD-S002)
	Alcohol Content	In-house Method (APIL-FOOD-E001)
II. Residues in foods and agricultural materials		
.01 Elements		
	Heavy metals: Arsenic	AAS/ICP-AES USEPA 6010
.02 Pesticides		
	Heavy metals: Arsenic	AAS/ICP-AES USEPA 6010
	Organochlorine	GC-ECD (AOAC 983.21, 970.52)
.03 Antibiotics		

	Organochlorine	GC-ECD (AOAC 983.21, 970.52)
.04 Insect infestation		
	Organochlorine	GC-ECD (AOAC 983.21, 970.52)

Product	Specific Test	Method Reference
III. Drugs and Pharmaceuticals		
.01 Drugs		
	Identification Tests	United States Pharmacopeia National Formulary (current version)
	Assay of active ingredients	
	Limit Tests	
	Residual Solvents	
.02 Herbal products		
	Identification Tests	United States Pharmacopeia National Formulary (current version)
	Assay of active ingredients	
	Limit Tests	
	Residual Solvents	
.03 Parenteral solutions and fluids		
	Identification Tests	United States Pharmacopeia National Formulary (current version)
	Assay of active ingredients	
	Limit Tests	
	Residual Solvents	
.04 Medicinal and veterinary preparations		
	Identification Tests	United States Pharmacopeia National Formulary (current version)
	Assay of active ingredients	
	Limit Tests	
	Residual Solvents	
.05 Vitamins		
	Dissolution Test	United States Pharmacopeia National Formulary (current version)
	Disintegration Test	
.06 Antibiotics		
	Dissolution Test	United States Pharmacopeia National Formulary (current version)
	Disintegration Test	

.11 Raw materials used in compounding pharmaceuticals		
	Dissolution Test	United States Pharmacopeia National Formulary (current version)
	Disintegration Test	

Product	Specific Test	Method Reference
IV. Cosmetics, perfumes and essential oils		
.01 Cosmetics		
	Heavy Metals: Lead, Cadmium, Arsenic	ASEAN Harmonized Method
	Hydroquinone	United States Pharmacopeia National Formulary (current version)
	Alcohol Content	In-House Method (APIL-COSM-I001)
	Viscosity	Brookfield Viscometer
	pH	Poucher
	Density	Pycnometer
.04 Hygiene products		
	Heavy Metals: Lead, Cadmium, Arsenic	ASEAN Harmonized Method
	Hydroquinone	United States Pharmacopeia National Formulary (current version)
	Alcohol Content	In-House Method (APIL-COSM-I001)
	Viscosity	Brookfield Viscometer
	pH	Poucher
	Density	Pycnometer
V. Detergents and related products		
.01 Soap		
	Heavy Metals: Lead, Cadmium, Mercury, Arsenic	ASEAN Harmonized Method
.02 Synthetic detergents		
	Heavy Metals: Lead, Cadmium, Mercury, Arsenic	ASEAN Harmonized Method
.05 Disinfectants		

	Heavy Metals: Lead, Cadmium, Mercury, Arsenic	ASEAN Harmonized Method
--	---	-------------------------

Product	Specific Test	Method Reference
VI. Water		
.01 Bottled water		
	pH	Electrometry (SMEWW 4500HB)
	Chloride	Titrimetry (SMEWW 4500 C1 B)
	Color	Visual (SMEWW 2120B) Spectrophotometry (SMEWW 2120C)
	Chromium (VI)	Spectrophotometry (SMEWW 3500Cr B)
	Nitrate	Spectrophotometry (ASTM D992.71)
	Nitrite	Spectrophotometry (SMEWW 4500NO ₂ B)
	Phosphate	Spectrophotometry (SMEWW 4500P D)
	Sulfate	Spectrophotometry (SMEWW 4500SO ₄ E)
	Turbidity	Nephelometry (SMEWW 2130B)
	Total Hardness	Titrimetry (SMEWW 2340 C)
	Total Dissolve Solids	Gravimetry (SMEWW 2540 C)
	Fluoride	APHA 2510 A & B
	Aluminum, Copper, Iron, Manganese	AAS (SMEWW 3111B)
	Heavy Metals: Lead, Cadmium, Chromium	AAS (SMEWW 3112B)
	Mercury	AAS (SMEWW 3114B) Cold Vapor AAS (SMEWW 3112B)
	Nickel	SMEWW 3111B, 3030F

Legend to Reference Methods:

AOAC - Association of Official Analytical Collaboration

SMEWW – Standard Methods for the Examination of Water and Wastewater

*****Nothing follows*****

Microbiological Testing

Product	Specific Test	Method Reference
I. Foods		
1.0 Milk and Dairy Products		
01.2 Sweetened Condensed Milk/ Creamer	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
01.4 Pasteurized milk	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
01.5 Pasteurized cream	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
01.6 Yogurt and other fermented milk	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
01.8 Processed cheese spread	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
01.9 Soft Cheese (from pasteurized milk)	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF, 5 th ed.)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
01.10 Hard and semi-hard cheese	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF, 5 th ed.)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
01.11 All raw milk cheese; raw milk unripened cheese with moisture > 50%, pH > 5.0	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
01.12 Cream cheese products	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
02.0 Fats, Oils, and Fat Emulsions		
02.1 Butter (whipped, pasteurized)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)

Product	Specific Test	Method Reference
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
02.2 Butter made from unpasteurized milk or unpasteurized milk products	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
02.3 Margarine	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Fecal Coliform	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
02.4 Virgin Coconut Oil	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
03.0 Edible Ices, Including Sherbet and Sorbet		
03.1 Ice cream and sherbet (plain and flavored)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
03.2 Ice cream with added ingredients (nuts, fruits, cocoa etc.)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
03.3 Flavored ice (e.g. Ice candy)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.0 Confectioneries		
04.1 Cocoa powder	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method /

Product	Specific Test	Method Reference
		MPN Method (FDA BAM Ch. 4)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.2 Chocolate products	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.3 Chocolate confectioneries (chocolate bars, blocks and bonbons)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.4 Sugar confectioneries (hard and soft candies, toffees, caramel, fondants, creams, nougats and pastes)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.5 Ethnic Milk-based Confectioneries (Pastillas and Yema)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
04.6 Ethnic Flour-based Confectioneries (Polvoron, Piaya and Barquillos)	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
04.7 Coconut Sap Sugar	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
04.8 Dry, Granulated Sugar	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
04.9 Liquid Sugar (sugar syrup)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)

Product	Specific Test	Method Reference
04.10 Non-nutritive sweeteners (e.g. aspartame, sucralose) and bulk sweeteners (e.g. polysols)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
05.0 Fruits and Vegetables, Nuts and Seeds		
05.2 Fermented vegetables, Ready to Eat (e.g. Kimchi)	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method / MPN Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
05.5 Coconut (desiccated)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
05.7 Sun dried fruits	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
05.8 Dried fruits (e.g. dried mangoes, dates, figs, apricots, raisins, etc.)	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
05.9 Ethnic food products – Sweet preserves in syrup e.g. Kaong, Langka, Banana, legumes, chickpeas, red beans, white kidney beans, Coconut, <i>Nata de Coco</i> , Mixed Preserves (halo-halo) <i>*Acidified sweet preserves: pH ≤4.6; (a_w) > 0.85</i>	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
05.10 Malunggay (Moringa) powder	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)

Product	Specific Test	Method Reference
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF 5 th ed.)
06.0 Egg and Egg Products		
06.1 Pasteurized egg products (smoked, liquid, frozen, or dried)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
07.0 Cereals and Cereal Products		
07.1 Breakfast cereals	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
07.2 Cereal/cereal grains	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
07.3 Cultured seeds and grains (Sprouted Seeds) e.g. bean sprouts, alfalfa etc.	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	Fecal Coliform	MPN Method (FDA BAM Ch. 4)
07.4 Soya flours, concentrates and isolates	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
07.5 Flour, corn meal, corn grits, semolina	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
07.6 Dry mixes (e.g. pancake mix, purple yam mix)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
07.8 Soy protein	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
07.9 Tofu	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method

Product	Specific Test	Method Reference
		(FDA BAM Ch. 12)
07.10 Pasta products and noodles Uncooked (wet and dry)	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
07.11 Starch	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
07.12 Coconut flour	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
07.13 Instant noodles (noodle cake)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
07.14 Instant noodles (mixed with seasoning)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
07.15 Rice based desserts (rice cakes), ready to eat e.g. kakanin, suman, layered, molded, rice balls, mochi	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
08.0 Bakery Products		
08.1 Frozen bakery products (ready to eat) with low acid or high a_w fillings or toppings	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)

Product	Specific Test	Method Reference
08.2 Frozen bakery products (to be cooked) with low acid or high a_w fillings or toppings (e.g. meat pies and pizzas)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
08.4 Baked goods (e.g. breads, cakes, pie crust, pastries and fried doughs)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
08.5 Coated or filled, dried shelf-stable biscuits	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
09.0 Ready to Eat Savouries		
09.1 Snack foods (e.g. potato chips, popcorn, pretzels, rice crackers, flavored crackers, processed whole nuts, yoghurt-cereal- and honey-covered nuts, dried fruit nut and cereal snacks, crackers with fish, fish product or flavoring)	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
10.0 Meat and Meat Products		
10.1 Dried animal products	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.2 Meat paste and paté (heat treated)	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.3 Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
10.4 Packaged cooked cured/salted meat (ham, bacon)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.5 Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages)	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.6 Cooked poultry meat, frozen to be re-heated before eating (e.g. prepared frozen meals chicken burgers, chicken turkey rolls, chicken nuggets, other breaded poultry meat products)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)

Product	Specific Test	Method Reference
10.7 Cured/smoked poultry meat	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.10 Marinated Meat Products e.g. Marinated meat and meat preparations (tapa, sisig, etc.), Marinated poultry, Dim sum made from meat (siomai)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
10.11 Minced meat and meat preparations made from poultry meat intended to be eaten cooked	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
10.12 Minced meat and meat preparations made from species other than poultry intended to be eaten cooked	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
10.13 Foods cooked immediately prior to sale or consumption e.g. Takeaway food, burgers, kebabs, sausages, pizza, ready meals (cook/chill and cook/freeze) after regeneration	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF 5 th ed.)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
11.0 Fish and Other Seafood Products		
11.1 Fresh frozen fish and cold-smoked	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.2 Pre-cooked breaded fish	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.3 Frozen raw crustaceans	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.4 Frozen cooked crustaceans	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)

Product	Specific Test	Method Reference
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.5 Cooked Crustaceans	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.6 Cooked, chilled and frozen crabmeat	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.7 Fresh and frozen bivalve molluscs	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
11.9 Ethnic food products – Dried, salted fish	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	MPN Method (FDA BAM Ch. 4)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
11.10 Smoked fish	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
11.11 Salt fermented fish and shrimps (bagoong)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
11.13 Aquatic products - cooked (heat-treated) aquatic product - intended for human consumption aquatic plant products (algae)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
11.14 Fish and crustacean based processed meat e.g. fish ball, squid ball	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
12.0 Spices, Soups, Sauces, Salads and Protein Products		
12.1 Dry mixes for soup and sauces	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)

Product	Specific Test	Method Reference
12.3 Spices	Molds Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
12.4 Spices (ready to eat)	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
	Molds Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
12.5 Food grade gelatin	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
12.6 Emulsified Sauce pH ≤4.6 (e.g. Mayonnaise, Thousand Island, Ranch, French)	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
12.7 Instant noodles seasoning	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
12.8 Ready to eat flavourings / instant condiments (e.g. soy sauce and sauce products, composite seasonings)	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
12.9 Aquatic dressing	<i>S. aureus</i>	Spread Plate Method (FDA BAM Ch. 12)
12.10 Salads and sandwich spreads (excluding cocoa milk-based sandwich spreads)	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
13.0 Beverages		
13.1 Non-alcoholic beverages (e.g. Ready to drink softdrinks, iced tea, energy drinks, jelly drinks)	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
13.2 Frozen juice concentrate	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
13.4 Powdered beverages (e.g. iced tea, powdered juices/mixes)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
13.5 Fruit beverage products	Aerobic Plate Count	Pour Plate Method

Product	Specific Test	Method Reference
		(FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
13.6 Bottled water (purified/distilled water)	Heterotrophic Plate Count	Pour Plate Method (SMEWW 9215B 24 th ed.)
	¹ Coliform Count	MTFT (SMEWW 9221B 24 th ed.)
13.7 Bottled water (mineral water)	Heterotrophic Plate Count	Pour Plate Method (SMEWW 9215B 24 th ed.)
	¹ Coliform Count	MTFT (SMEWW 9221B 24 th ed.)
13.8 Chilled young coconut water (buko juice)	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Yeast and Mold Count	Pour Plate Method / Spread Plate Method (FDA BAM Ch. 18)
	Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
14.0 Food for infants and young children		
14.1 Powdered infant formula with or without added lactic acid producing cultures (intended for 0 to 6 months old) Formula for medical purposes and human milk fortifiers	<i>E. coli</i>	MPN Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF, 5 th ed.)
	Coliform Count	MPN Method (FDA BAM Ch. 4)
14.2 Follow up formula /milk supplement (from 6 months infants to 36 months young children)	<i>E. coli</i>	MPN Method (FDA BAM)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	<i>Enterobacteriaceae</i>	Pour Plate Method (CMMEF, 5 th ed.)
	Coliform Count	MPN Method (FDA BAM Ch. 4)
14.5 Dried and instant products requiring reconstitution	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	¹ Coliform Count	MPN Method (FDA BAM Ch. 4)
14.6 Dried products requiring reconstitution and boiling before consumption	¹ Coliform Count	Pour Plate Method (FDA BAM Ch. 4)
	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
14.7 Cereal based foods for infants	Aerobic Plate Count	Pour Plate Method (FDA BAM Ch. 3)
	Coliform Count	MPN Method (FDA BAM Ch. 4)
II. Drugs and Pharmaceuticals		
01.0 Drugs 02.0 Herbal Products 03.0 Medicinal Preparations	Total Yeasts and Molds Count	United States Pharmacopeia National Formulary (USP NF) (Current version)
	<i>Enterobacteriaceae</i> Count	
	<i>Escherichia coli</i>	

Product	Specific Test	Method Reference
04.0 Parenteral solutions and fluids	<i>Staphylococcus aureus</i>	
III. Cosmetics, Perfumes, Hygiene Products and Household Hazardous		
01.0 Cosmetics 02.0 Perfumes 03.0 Hygiene Products	Yeast and Mold Count	Spread Plate Method (ASEAN Harmonized Method for Cosmetics)
04.0 Detergents and other Products	<i>Staphylococcus aureus</i>	FDA BAM, Chapter 23
	<i>Pseudomonas aeruginosa</i>	FDA BAM, Chapter 23

¹If positive for Coliform, *E. coli* must be tested and should not be detected.

Legend to Reference Methods

BAM - Bacteriological Analytical Manual

SMEWW - Standard Methods for the Examination of Water and Wastewater

CMMEF - Compendium of Methods for the Microbiological Examination of Foods

***** Nothing Follows *****